Mariano Marcos State University
SDG 15 · Life on Land

STORYBOARD · CHAPTER 03 OF 13

03

Maintaining and extending biodiversity

9 frames

15 LIFE ON LAND

THE STORY

Three ecosystems, one method: raise the stock locally, plant it with partners and protect what survives: forest, food plants and mangrove.

MMSU'S COMMITMENT

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

THE FILMSTRIP · 9 FRAMES

2025 Ongoing programme


1


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7


8

NO PHOTO
9

AT A GLANCE

1,068.13 ha
forest reserves

46
food plants documented

280
mangrove propagules (Dec)



Bamboo planting site in the University Forest Reserve, Brgy. Mabaleng, 2 July 2025. (Photo: MMSU STRATCOM)

FOREST RESERVE AREAS · BARANGAYS LUBBOT AND PAYAO, CITY OF BATAC

Enriching MMSU's 1,068-Hectare Forest Reserves with Native Species

1. **THE LAND**
   MMSU manages Forest Reserve Areas in Lubbot and Payao totalling 1,068.13 hectares, where its forestry and agroforestry research is set up.
2. **OWN PLANTING STOCK**
   The university maintains and extends these ecosystems with seedlings and propagules raised in its own MMSU Central Nursery and MMSU Bamboo Nursery.
3. **2025 ENRICHMENT**
   2 Jul 2025: 230 CAFSD graduating students added 120 guijo and yakal seedlings in Payao and 130 kawayan tinik, machiku and giant bamboo propagules in Mabaleng.

1,068.13 ha
managed forest reserves

250
plants added in 2025

Source: MMSU Research Directorate, Station Profile · MMSU News, "CAFSD graduands plant trees, bamboos in Batac forest reserves," 4 July 2025



Partners receive indigenous food plant planting materials for the Community Foodscapes and Seedbanks, World Food Day 2025.

INDIGENOUS FOOD PLANTS PROGRAM · RESEARCH AND EXTENSION DIRECTORATES

Conserving Ilocos Indigenous Food Plants In Situ, Ex Situ and Circa Situ

1. **THREE WAYS OF CONSERVING**
   An IFP Botanical Garden (ex situ), a himbabao (alukon) genebank at the MMSU Forest Reserve, circa situ cultivation of the wild yams kamangeg and dolian, and integration of IFPs into community foodscapes.
2. **OCTOBER 2025**
   For World Food Day, the second batch of Community Foodscapes and Seedbanks was set up with LGUs and schools at Eladio V. Barangan Memorial Elementary School, San Nicolas, joining sites in Sarrat and Batac.
3. **RECOGNITION**
   First R&D initiative adopted by the Regional Development Council of Region 1. A study by Dr. Menisa A. Antonio et al. won Best Paper at the SUC-ACAP 2025 Biennial Conference (27–29 Aug, Bohol).

46	7	2nd
food plants documented	towns	batch of foodscapes

Source: MMSU Research Directorate, “MMSU research on indigenous food plants wins best paper at SUC-ACAP 2025 Conference,” 29 August 2025 · MMSU Extension Directorate, “MMSU expands indigenous food plant seedbanks and foodscapes in LGUs,” October 2025 · MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking



Mangrove site, from the gallery of the October 2025 seminar-workshop of the NbS mangrove project. (Photo: MMSU)



Dr. Arlene L. Gonzales at the MMSU–DENR Region 1 dialogue on mangrove conservation, ILAARRDEC.

THROUGHOUT 2025 · BADOC AND CURRIMAO, ILOCOS NORTE

Nature-based Solutions for Mangrove Ecosystems in Badoc and Currimao

1. **THE PROJECT**
   DOST-PCAARRD-funded nature-based solutions project led by Dr. Arlene L. Gonzales, implemented with DENR-PENRO Ilocos Norte, the Provincial Government and the LGUs of Badoc and Currimao.

2. **THE METHOD**
   Green engineering with locally available, ecologically friendly materials, including bamboo-engineered structures that protect young mangroves and improve their survival.

3. **2025 MILESTONES**
   An MMSU–DENR Region 1 dialogue at ILAARRDEC (DENR pledged technical support in restoration, monitoring and evaluation) and the 27–28 Oct propagation seminar-workshop for four coastal towns.

2
project municipalities

4
towns trained in 2025

Source: MMSU News, “MMSU trains Ilocos Norte coastal towns on mangrove propagation and management,” 5 November 2025 · MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking



MMSU, Rotary Club of Laoag Sunshine and LGU-Badoc plant mangrove propagules in Brgy. Pagsanahan, Badoc, 6 December 2025. (Photo: MMSU)



Participants of the mangrove enhancement and growing activity in Badoc, 6 December 2025. (Photo: MMSU)

6 DEC 2025 · BRGY. PAGSANAHAN, BADOC, ILOCOS NORTE

MMSU, Rotary Club and Badoc LGU Restore a Typhoon-Damaged Mangrove Stand

1. **THE PLANTING**
   MMSU, the Rotary Club of Laoag Sunshine and LGU-Badoc planted 280 mangrove propagules to restore a stand extensively damaged by past typhoons.
2. **THE GOAL**
   To ensure the continuity of ecological succession at the site.
3. **WHO LED IT**
   President Manzano with Badoc's Hon. Virgilio M. Calajate and Rotary's Atty. Zsa Zsa Raval; MMSU's Vice President for Resource Generation and directors for Extension, Curriculum Development and Innovative Learning, and Student Affairs also took part.

280
mangrove propagules

3
partners

Source: MMSU News, "MMSU ties up with Rotary Club of Laoag Sunshine for mangrove planting in Badoc town," 11 December 2025



Participants of the bamboo propagation, nursery and plantation management training, Brgy. Aglipay, Pinili, 2025. (Photo: MMSU)

2025 · BRGY. AGLIPAY, PINILI, ILOCOS NORTE

A New Bambusetum in Barangay Aglipay, Pinili

1. **THE TRAINING**
   In a two-day bamboo production training, the MMSU Extension Directorate and CAFSD worked with Barangay Aglipay and ILAARRDEC.
2. **THE RESULT**
   46 farmers, local leaders and community members established a new bambusetum, a collection of bamboo species, in the barangay.
3. **THE SKILLS**
   Bamboo propagation, nursery establishment and plantation management.

46	2	1
participants	training days	new bambusetum

Source: MMSU News, “Sustainable livelihood: MMSU, ILAARRDEC hold bamboo training in Pinili,” 3 September 2025



Capture: Philippine News Agency post of 6 June 2026 on indigenous food plants propagated at the MMSU NRCMC botanical garden (photos: Leilanie Adriano, PNA).



Partners receive indigenous food plant planting materials for the Community Foodscapes and Seedbanks, World Food Day 2025.

6 JUN 2026 · MMSU NATURAL RESOURCES CONSERVATION AND MANAGEMENT CENTER, BATAC

Indigenous Food Plants Propagated at the MMSU Botanical Garden Are Shared with Schools and Communities

Ongoing programme · Philippine News Agency, 6 June 2026

1. **THE REPORT**
   The Philippine News Agency reported that sugod-sugod, alokon, buga and karot are being propagated at the NRCMC botanical garden.
2. **THE SHARING**
   Planting materials go to schools and community stakeholders for food gardens, to boost food security and give farmers an additional income source.
3. **THE LINK**
   The garden is the ex situ arm of the IFP Program, which in 2025 extended community foodscapes and seedbanks to new partner barangays.

Source: Philippine News Agency, Facebook post on indigenous food plants at the MMSU Natural Resources Conservation and Management Center botanical garden, 6 June 2026



Project Leader Dr. Arlene L. Gonzales presents the project at its launch, MMSU Review Center, 27 June 2024. (Image credit: TTPD, DOST-PCAARRD)



The MMSU project team with DOST-PCAARRD, DENR-PENRO Ilocos Norte, the Provincial Government of Ilocos Norte and the LGUs of Badoc and Currimao at the launch. (Image credit: TTPD, DOST-PCAARRD)

1 MAY 2024 – 30 APR 2026 · BADOC AND CURRIMAO

Mangrove Nature-based Solutions Project Runs from 2024 to 2026

Ongoing programme · 1 May 2024 to 30 April 2026

1 **THE PROJECT**
The DOST-PCAARRD-funded NbS project led by Dr. Arlene L. Gonzales launched on 27 Jun 2024 at the MMSU Review Center, supported by DENR-PENRO Ilocos Norte, the Provincial Government and the LGUs of Badoc and Currimao.

2 **THE PROBLEM**
Poor hydrological conditions keep young mangrove propagules from surviving.

3 **THE FIX**
Technology-assisted plantations using locally available, ecologically friendly materials, including bamboo-engineered structures.

24	2
months of project term	municipalities

Source: DOST-PCAARRD, “DOST-PCAARRD, MMSU launch nature-based solutions project on mangrove management in Ilocos Norte,” 7 August 2024



Launch of the bambusetum at the MMSU Dingras Campus, 30 September 2026. (Photo: MMSU STRATCOM)

SEP 2025 · 30 SEP 2026 · MMSU DINGRAS CAMPUS

Bambusetums Create Living Collections for Bamboo Conservation and Learning

Ongoing programme · September 2025 and September 2026

1 **THE LAUNCH**
During Bamboo Month, the MMSU Dingras Campus, through CAFSD, launched its bambusetum on 30 Sep 2026, the second in the university after the Main Campus.

2 **WHO TOOK PART**
Students, faculty, field workers and stakeholders planted with For. Charlie Maine Cacactin; four student and staff organisations supported the activity.

3 **THE PURPOSE**
A living collection of bamboo species and learning site for teaching, research, technology development and extension. It follows the Aglipay, Pinili bambusetum of 2025.

2nd
campus bambusetum

Source: MMSU News, “MMSU CAFSD-Dingras launches bambusetum to advance bamboo research and education,” 2 October 2026 · MMSU News, “Sustainable livelihood: MMSU, ILAARRDEC hold bamboo training in Pinili,” 3 September 2025

13 Jan 2023 · Biodiversity Conservation Learning Center and Leisure Park, Main Campus

28
native tree species

120
seedlings (2023)

2017
Green Wall began

13 JAN 2023 · BIODIVERSITY CONSERVATION LEARNING CENTER AND LEISURE PARK, MAIN CAMPUS

Biodiversity Conservation Learning Center and Leisure Park Is Enriched with 28 Native Tree Species

Ongoing programme · since 2023

1. **THE PLANTING**
120 more seedlings of 28 native tree species and two bamboo species were planted, led by then President Shirley C. Agrupis with administrators, faculty, staff and students.

2. **HOW IT IS USED**
A field laboratory for classes, identified for development into a health and wellness tourism hub endorsed by the provincial government.

3. **THE TRADITION**
Continues planting begun in 2017 with the Green Wall project on the Batac campus, extending to Nagbacsan, Currimao, Dingras and Laoag, in support of the provincial Green Wall of Ilocos Norte and DENR's National Greening Program.

28
native tree species

120
seedlings (2023)

2017
Green Wall began

Source: MMSU News, "More trees planted at MMSU to establish 'health tourism' hub," 13 January 2023

WHERE THE STORY LANDS

Forest reserves, indigenous food plants, mangroves and bamboo: MMSU raises planting stock in its own nurseries, plants with partners, and keeps collections and seedbanks that schools and communities can draw on.

1,068.13 ha
forest reserves

46
food plants documented

280
mangrove propagules (Dec)

THE FRAMES

1. **Enriching MMSU's 1,068-Hectare Forest Reserves with Native Species**
   Forest Reserve Areas
2. **Conserving Ilocos Indigenous Food Plants In Situ, Ex Situ and Circa Situ**
   Indigenous Food Plants Program
3. **Nature-based Solutions for Mangrove Ecosystems in Badoc an...**
   Throughout 2025
4. **MMSU, Rotary Club and Badoc LGU Restore a Typhoon-Damaged...**
   6 Dec 2025
5. **A New Bambusetum in Barangay Aglipay, Pinili**
   2025
6. **Indigenous Food Plants Propagated at the MMSU Botanical Garden Ar...**
   6 Jun 2026
7. **Mangrove Nature-based Solutions Project Runs from 2024 to 2026**
   1 May 2024 – 30 Apr 2026
8. **Bambusetums Create Living Collections for Bamboo...**
   Sep 2025
9. **Biodiversity Conservation Learning Center and Leisure Park Is Enrich...**
   13 Jan 2023

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